



I am designing a planning tool that links an asteroid mission planner to a rotating momentum-exchange tether (rotovator) in Earth orbit. The tether's center of mass is in a 610 km circular orbit. It has a 320 km half-length and a 2.5 km/s tip speed, so the upper tip moves at about 10.06 km/s inertial at a 7,301 km radius. That is below local escape speed. Payloads returning from near-Earth asteroids arrive on hyperbolic trajectories with an Earth-arrival v -infinity. The tether would catch them at the upper tip after a small matching burn, then either release them at the lower tip for Earth entry, or keep them in orbit. Each inbound catch puts orbital energy back into the tether, offsetting outbound throws (the "balanced flow" idea from MXER studies).

Answer, with primary sources (NASA/NTRS technical reports, AIAA/JBIS papers, Tethers Unlimited and Boeing study reports, peer-reviewed papers):

1. What Earth-arrival v -infinity and asymptote-declination ranges do published studies find for returns from accessible near-Earth asteroids (e.g., NHATS targets, ARM/ARRM studies, Keck Institute 2012 asteroid retrieval study, sample-return missions such as OSIRIS-REx and Hayabusa2)? Give per-mission or per-target values where published.
2. What catch relative-velocity tolerance, position tolerance and timing window did MXER, HASTOL or other rotovator rendezvous studies assume or demonstrate, and on what basis?
3. In MXER-class designs, what was the total facility mass (tether + central station + ballast), and what payload-to-facility mass ratio did they assume? How did they handle the perigee drop after an unbalanced throw?
- 4.

How have published studies modeled inbound "catch and drop" or "catch and hold" traffic restoring tether momentum? Did any quantify the fraction of reboost demand that return traffic offsets?5. For returning payloads, how does aerocapture or direct entry (aeroshell mass fraction, peak heating, peak g) compare with propulsive capture at v-infinity of 1–5 km/s? Cite heat-shield mass-fraction data.

FOLLOW-UP CHAIN DIRECTIVE:After answering the question above in full, continue as follows:

LEVEL 1 — State the 3 most decision-relevant follow-up questions your answer raises for a planning tool that decides whether an asteroid return payload should be caught by a LEO rotovator, and answer each with sources.

LEVEL 2 — For each Level-1 answer that materially affects a design decision, pose and answer the single most important follow-up it raises, with sources.

LEVEL 3 — Repeat once more for any Level-2 answer that still carries open decision weight.

BUDGET: no more than 10 follow-up answers total across all levels. Prune by decision-relevance, not curiosity — drop branches that only add color.

For EVERY follow-up answer:(a) open with one line stating why this follow-up matters for the tool,(b) cite primary sources,(c) flag each number as official-published vs third-party-estimated.

END with a section titled **LOAD-BEARING NUMBERS:** a flat list of every number in this entire response that a design decision might rest on — one line per number, with its source. This list feeds an independent verification pass.

Your rotovalor can accept asteroid-return traffic only after a substantial velocity-matching maneuver: even a parabolic arrival cannot match its upper tip without braking. Published tether studies support momentum recovery from deliberately designed reverse traffic, but they do not establish that arbitrary asteroid arrivals, or payloads merely caught and retained, automatically balance outbound throws. ^[1]

The strongest verified numbers and their limitations follow. Several requested quantities—especially mission-specific arrival-asymptote declinations, a flight-validated catch-error envelope, and an asteroid-return percentage of reboost savings—are not established by the primary sources retrieved here. I distinguish those gaps from published results rather than supplying assumed ranges.

Provenance: “Published” means reported by NASA/JPL, a study contractor, or mission authors; it does not necessarily mean flight-demonstrated. “Estimated” means my calculation or interpretation, not an official mission result. Your tether parameters are design inputs, not published specifications.

1. Asteroid arrival conditions

Published retrieval trajectories

The Keck study’s March 2012 presentation contains a particularly useful target-by-target table: slide 31 reports Earth-arrival C_3 , returned mass, and trajectory duration. For positive Earth-arrival C_3 , the corresponding two-body excess speed is $v_\infty = \sqrt{C_3}$. These trajectories target lunar-assisted capture, not rendezvous with a LEO tether. ^[2]

Target and trajectory variant	Returned mass, published	Arrival C_3 , published	Corresponding v_∞ , estimated
2008 HU4	250 t	1.8 km ² /s ²	1.342 km/s
2008 HU4	400 t	1.7 km ² /s ²	1.304 km/s
2008 HU4	650, 950, or 1,300 t variants	1.6 km ² /s ²	1.265 km/s
1998 KY26	30 or 60 t variants	2.0 km ² /s ²	1.414 km/s
2000 SG344	1,800 t	2.0 km ² /s ²	1.414 km/s
2000 SG344	3,600 t	2.1 km ² /s ²	1.449 km/s

All published entries in this table come from the study presentation; the square roots are calculated here. ^[2]

The same table includes 2008 HU4 and 2000 SG344 variants with arrival $C_3 = 0.0$, explicitly identified as capture to Sun–Earth L2 through a stable manifold. Do not translate those into an ordinary parabolic LEO arrival: the destination and dynamical model differ. ^[2]

The April 2012 final report likewise selects high lunar orbit rather than LEO, partly because transporting the captured mass deeper into Earth’s gravity well requires more propulsion and because destination safety matters. Its return architecture uses SEP and lunar gravity assists. ^[3]

NASA’s subsequent ARM study uses a multi-revolution Lambert scan with lunar-gravity-assist performance limits dependent on departure and arrival declinations. However, the retrieved paper does not publish a usable target-by-target table of arrival v_∞ and declination. Its return-mass results are not substitutes for those boundary conditions, and this early ARM paper should not be presented as the final ARRM reference trajectory. ^[4]

NHATS: useful, but easy to misread

NHATS explicitly publishes a “Return Declination” field for individual trajectory solutions. It also distinguishes entry speed from “Earth return dV”: the latter is the braking required to satisfy the atmospheric-entry-speed constraint, not arrival v_∞ . ^{[5] [6]}

The published NHATS constraint is entry speed no greater than 12 km/s at 125 km altitude. In a nonrotating, two-body interpretation, an *unbraked* trajectory satisfying this limit has $v_\infty \lesssim 4.63$ km/s—an estimate here, not an NHATS-published population range. Trajectories requiring return braking invalidate the interpretation of that ceiling as their original arrival excess speed. ^[7]

Consequently:

- NHATS accessibility does not establish tether accessibility.
- Its departure C_3 must not be used as return C_3 .
- Return declination must be associated with the particular departure date, flight time, stay time, and trajectory solution—not assigned permanently to the asteroid. ^{[6] [5]}

I did not retrieve a verified NHATS population-wide arrival-declination range or numerical per-target declinations. A defensible planner should ingest solution records rather than use an unsupported generic declination band.

Sample-return missions

Mission	Published return condition	What it establishes
OSIRIS-REx, early design	Nominal entry velocity 12.2 km/s and entry flight-path angle -8.2°	A capsule-entry design point; the flight-path angle is not asymptote declination. ^[8]
OSIRIS-REx, operational return paper	Atmosphere-relative entry velocity 12.38 km/s; entry-interface radius 6,503.142 km	A more relevant operational value, but not directly an inertial v_∞ . ^[9]
Hayabusa2	Approximately 11.6 km/s ground-relative/geodetic entry velocity	Demonstrates direct interplanetary entry; Earth rotation must be accounted for before deriving inertial arrival energy. ^[10]

Treating the early OSIRIS-REx value of 12.2 km/s as inertial at the operational interface radius would give approximately 5.12 km/s excess speed. That is only an illustrative estimate mixing an early design speed with an operational radius—not a published or reconstructed mission v_∞ . ^{[8] [9]}

For Hayabusa2, I would not give an “exact” excess speed from the published ground-relative speed alone. Both missions require the inertial entry state or arrival ephemeris to obtain an

auditable excess-speed vector and asymptote declination.

2. Tether operations evidence

Catch tolerances and demonstration basis

The NASA-supported Quadtrap work is stronger evidence for mechanism feasibility than for an operational error envelope.

Quantity	Verified published result	Appropriate interpretation
Nominal relative velocity	Tether and payload orbits are selected to match position and velocity instantaneously	Nominal rendezvous condition, not permission to absorb a large unmatched arrival speed. ^[11]
Position-error geometry	Approximately 10:1 major-to-minor error-ellipse ratio	Derived from Monte Carlo tether-dynamics analysis using available data or reasonable input estimates. ^[11]
Capture duration	Closure must occur within 5 s or less	A mechanism requirement based on motion through the proposed error window, not a symmetric arrival-time tolerance. ^[11]
Representative closure design	3.5 s closure with support-cable length no greater than approximately 100 m under approximately 2 g	A mechanism-sizing example; the cable length is not an asserted position tolerance. ^[11]
Prototype	Approximately 1:10 dimensional scale	Ground proof of concept, not an orbital demonstration. ^[11]
Prototype aperture	1 m square	Physical prototype dimensions, not a demonstrated full-scale capture envelope. ^[11]
Prototype passage window	Up to 0.9 s	Laboratory window associated with its probe and test trajectory. ^[11]
HASTOL timing	Capture window shorter than 10 s	Concept-study requirement; the presentation explicitly says simulation methods still need experimental validation. ^[12]

The Quadtrap paper explicitly warns against inferring validation at other scales from the single prototype comparison. It also states that guaranteed closure requires consideration of the momentum associated with velocity error and probe deformation; its closure-time rules alone do not establish that capacity. ^[11]

Missing: I could not verify a single operationally qualified tolerance such as “ $\pm X$ metres, $\pm Y$ m/s, $\pm Z$ seconds” for this asteroid-return application. Do not substitute the error-ellipse aspect ratio, mechanism dimensions, or nominal zero-relative-speed condition for such a specification.

Facility mass and payload ratio

The Tethers Unlimited/Boeing presentation provides an unusually clear mass breakdown for its initial LEO–GTO facility. ^[13]

Component or configuration	Published mass
Control station, current-best estimate without margin	10,967 kg
Tether	8,274 kg
Grapple	650 kg
Retained expended upper stage	3,467 kg
Total on-orbit facility	23,358 kg
Initial payload	2,500 kg

Those component values sum to the published on-orbit total. The same presentation separately lists the control-station mass with contingency; mixing that value into the current-best-estimate total would create an inconsistent mass budget. ^[13]

The corresponding estimated payload-to-facility mass ratio is **0.107**, or facility-to-payload ratio **9.34**. ^[13]

Other published configurations differ:

- A larger multi-destination point design has **63 t launch mass**, handles **5 t** lunar/high-energy payloads or **10 t** GTO payloads, retains launch stages for ballast, and specifies **200 kW** end-of-life power for approximately **one month** of reboost. Launch mass is not necessarily the final all-in on-orbit mass including retained stages. ^[13]
- The earlier cislunar Earth facility has published mass **26,250 kg** for a **2,500 kg** payload, giving a facility-to-payload ratio of **10.5**. ^[1]

These are architecture-specific point designs, not a universal MXER mass ratio. They also do not validate scaling the much longer tether in your proposal with the same ratio.

Unbalanced throws and perigee

The studies do not avoid recoil; they explicitly allocate facility orbital energy and momentum to the payload, then restore the facility through electrodynamic propulsion. The cislunar design uses an elliptical Earth orbit and electrodynamic reboost near perigee. Its simulation reports restoring the orbit within **85 days with 11 kW**, while the architecture summary quotes **95 days** between lunar boosts. These are different operational quantities, not contradictory values to average together. ^[1]

The later LEO–GTO presentation includes an **operational-orbit lifetime requirement of 15 days** and a **30-day turnaround requirement**. Thus, survival of the depleted orbit was explicitly an operational constraint. ^[13]

For your planner, the relevant constraint is the altitude of the entire rotating tether after recoil—not just COM perigee. Facility mass, payload mass, spin change, shifted COM, elastic motion, and tether orientation must enter the post-event propagation.

Balanced inbound traffic

The 1999 Tethers Unlimited report models a two-way cislunar architecture: an Earth tether throws material toward the Moon; a lunar tether catches and deposits it, picks up return material, and sends that material to the Earth tether for deboosting into LEO. The report states that equal reverse-flow masses can conserve system orbital momentum and energy in the designed architecture. Its simulations include three-dimensional orbital mechanics, bead-and-spring tether dynamics, and electrodynamic interaction.^[1]

This supports **ideal cancellation of transport-induced reboost** for properly matched reverse traffic. It does not establish cancellation of drag, disturbances, navigation corrections, or all operational energy consumption.^[1]

I did not find a verified study result such as “asteroid-return traffic offsets a specified percentage of annual MXER reboost.” The defensible published endpoint is ideal balanced reverse transport—not an empirically established utilization fraction.

For catch-and-hold, an important distinction remains: zero-relative-speed attachment by itself does not extract a payload’s hyperbolic excess energy. Your pre-catch burn removes some energy; subsequent carrying, reeling, and release determine what the facility gains. Retaining cargo changes facility mass and inertia and cannot be credited as the reverse of an outbound throw without modeling the final state.

3. Capture versus atmospheric arrival

Your configuration’s matching requirement

Using your radii and tip speed, and a two-body Earth model:

$$v_{\text{arrival}}(r_u) = \sqrt{v_{\infty}^2 + \frac{2\mu_E}{r_u}}, \quad \Delta v_{\text{match},\min} = v_{\text{arrival}} - v_{\text{upper}}.$$

This is a lower bound requiring favorable, parallel velocity vectors. The actual burn is

$$\Delta v_{\text{match}} = \|\mathbf{v}_{\text{arrival}} - \mathbf{v}_{\text{tip}}\|.$$

All values below are **estimated**, not published mission results. The transport/reboost interpretation follows the published tether energy-exchange model.^[1]

Arrival v_{∞} , km/s	Minimum upper-tip matching burn, km/s	Direct circular capture at upper-tip radius, km/s	Direct-entry inertial speed at interface, km/s	Circular-capture propellant fraction at assumed 450 s I_{sp}
1	0.441	3.108	11.117	50.6%
2	0.583	3.250	11.251	52.1%
3	0.815	3.483	11.471	54.6%
4	1.133	3.800	11.772	57.7%
5	1.528	4.195	12.149	61.4%

Calculations use your upper-tip radius of 7,301 km and NASA’s OSIRIS-REx interface radius of 6,503.142 km; the latter is only a consistent comparison surface, not a universal entry-interface definition. Propellant fractions exclude tanks, engines, margins, and dry capture hardware. The 450 s specific impulse is an analyst-selected comparison assumption. ^[9]

At your upper tip, estimated local escape speed is 10.449 km/s, versus tip speed 10.056 km/s. Thus even $v_{\infty} = 0$ requires approximately 0.393 km/s of favorable-direction braking. There is no positive arrival excess speed that matches this tip naturally.

Also, “propulsive capture” must name its final orbit. Barely binding the payload into a very high ellipse costs much less than circularizing it; the table deliberately compares circular capture, not minimum capture into any bound orbit.

Catch-and-drop energy

In the negligible-payload-mass, unchanged-tether approximation, the upper and lower tip speeds are approximately 10.056 and 5.056 km/s. Their specific orbital energies imply approximately 43.03 MJ/kg transferred from payload to facility during upper-to-lower carrying and release. This is an estimate of the *carry-and-drop* exchange, not energy delivered by the velocity-matched attachment itself. ^[11]

The corresponding lower-tip release would reach the selected atmospheric interface at approximately 5.336 km/s inertial. Its osculating trajectory intersects Earth, so it is an entry trajectory rather than a holding orbit. These estimates omit facility recoil, elasticity, atmosphere-relative velocity, and release targeting.

This can substantially reduce entry speed, but **peak heating and peak g cannot be determined from v_{∞} alone**. They also depend on ballistic coefficient, nose radius, flight-path angle, lift/drag, atmospheric density, and guidance. NASA’s Earth-entry TPS study explicitly varies entry speed, flight-path angle, and ballistic coefficient rather than assigning one heat-shield fraction to a speed. ^[14]

Heat-shield mass and loads

Primary-source example	Published number	What you can use it for
Stardust	Heatshield TPS approximately 22% of entry mass	Earth-return TPS heritage; not total aeroshell, parachute, avionics, or recovery-system fraction. ^[15]
Genesis	Heatshield TPS approximately 18% of entry mass	Another Earth-return architecture reference, not a universal aerocapture coefficient. ^[15]
Apollo	Heatshield TPS approximately 14% of entry mass	Earth-return heritage at different scale and geometry. ^[15]
Stardust	Entry approximately 12.5 km/s , peak stagnation heating approximately 1,200 W/cm²	Published high-speed Earth-return reference. ^[15]
Hayabusa2	Maximum heat flux approximately 12 MW/m² , equivalent to 1,200 W/cm²	Published capsule thermal environment; not scalable directly to bulk cargo. ^[10]

Primary-source example	Published number	What you can use it for
OSIRIS-REx	Survivability limits: 1,051 W/cm ² , 30.3 kJ/cm ² , 40 g	Release-acceptance limits, not measured nominal peaks; required for 99% of Monte Carlo cases. ^[9]

For OSIRIS-REx, the operational paper also distinguishes the physical heating limits from reduced limits used with POST2's convective-only heating model. That distinction matters: comparing a convective-only prediction directly against a combined-heating limit can incorrectly certify an entry. ^[9]

NASA's Sepka-Samareh Earth-entry TPS estimating study covers published ranges of 10–16 km/s, absolute flight-path angles 5–25°, and ballistic coefficients 41.95–128.74 kg/m². Its PICA correlation has a published 1,200 W/cm² cold-wall heating limit, 6.3% standard-deviation agreement with FIAT, and 11.7% maximum underprediction within its evaluated dataset. These are correlation properties, not guaranteed vehicle margins. ^[14]

For your tool:

- Direct entry delivers material to Earth and discards orbital energy.
- Aerocapture retains material in orbit but requires an atmospheric exit target and subsequent orbit management.
- Propulsive capture avoids atmospheric exposure but incurs propellant and propulsion-system mass.
- Tether catch-and-drop adds rendezvous and facility risks but can recover transport energy and reduce atmospheric-entry speed.

The verified Earth-return TPS fractions above are appropriate sanity checks. They are **not** sufficient to assert that an Earth aerocapture system for your cargo will weigh the same fraction. I did not retrieve a directly applicable, verified Earth-aerocapture aeroshell fraction across your full excess-speed range.

The attached calculation CSV contains the estimated screening table.

4. Decision-relevant follow-ups

The following chain contains nine answers: three at each level. Numerical values retain the provenance labels above.

Level 1 — A: Is matching affordable?

Why this matters: the tool must reject geometrically impossible or propulsion-infeasible catches before assigning momentum-recovery value.

Answer: require a propagated Earth-arrival state, not just scalar excess speed and declination. Solve for the interception epoch, tip position, tip velocity, and matching burn together. The published MXER mechanism starts from intentionally matched nominal position and velocity; NHATS supplies trajectory-specific return information, not an automatic tether rendezvous. ^[5]

For your configuration, the estimated matching-burn lower bounds in the comparison table already show that the burn is not merely a navigation correction.

Level 1 — B: Does the return restore usable momentum?

Why this matters: an energy credit that ignores spin or orbital geometry can falsely classify traffic as balanced.

Answer: compute changes in facility COM orbit, spin angular momentum, and energy through the complete maneuver. The published cislunar balance concerns reverse transport between specified endpoints, not arbitrary inbound cargo attachment. Catch-and-drop can be credited from its modeled final state; catch-and-hold requires explicit storage and mass-distribution modeling. ^[1]

Level 1 — C: Which destination is required?

Why this matters: Earth delivery and orbital delivery are different services and must not share one capture-cost metric.

Answer: compare only architectures reaching the requested destination. For Earth delivery, compare direct entry against catch-and-drop. For orbital delivery, compare aerocapture, propulsive capture, and tether-assisted placement into the specified usable orbit. The Keck study selected lunar storage rather than LEO precisely because propulsion and safety depend on destination. NASA defines aerocapture as a guided atmospheric pass establishing a bound orbit, not landing. ^{[16] [3]}

Level 2 — A: What arrival geometry is sufficient?

Why this matters: declination-only screening can accept trajectories whose local velocity vectors cannot match the tip.

Answer: retain the full incoming excess-velocity vector, encounter epoch, B-plane targeting, and inertial reference frame. Arrival asymptote declination is not local flight-path angle or encounter-orbit inclination. OSIRIS-REx's operational targeting uses B-plane states, time-of-arrival information, and propagated entry states—not a declination scalar alone. ^[9]

No numerical declination acceptance band is justified until you specify tether orbit inclination, node, spin plane, and maneuver authority.

Level 2 — B: How much can arrive before control limits fail?

Why this matters: aggregate inbound balance cannot prevent an individual event from violating altitude or structural constraints.

Answer: enforce per-event limits on recoil, minimum tether altitude, tension, spin, and reboost availability. Use a finite facility mass and inertia; the cislunar study explicitly moved beyond earlier infinite-ballast assumptions. Published facility-to-payload ratios are starting points for reproducing their point designs, not a safety certification for your geometry. ^{[13] [1]}

Level 2 — C: Can thermal mass be screened cheaply?

Why this matters: a planner needs fast mass estimates without disguising them as certified entry designs.

Answer: propagate a simplified atmospheric trajectory, integrate convective and radiative heating, then apply a published TPS estimating relationship only within its validated regime. Sepka and Samareh explicitly describe their correlations as early feasibility tools and still require high-fidelity analysis for a proposed mission.^[14]

Your estimated lower-tip entry speed lies below that paper’s published velocity range; do not extrapolate its correlation silently.

Level 3 — A: What makes rendezvous robust?

Why this matters: a nominal intersection is not a catch probability.

Answer: propagate joint arrival, facility, tip-dynamics, and maneuver-execution uncertainty into relative position and velocity. Evaluate the actual mechanism envelope and missed-catch outcome. The Quadtrap error-space work uses Monte Carlo dynamics; OSIRIS-REx similarly evaluates dispersed propagated states before committing to release. Neither establishes flight-qualified rotovator capture probability.^[17] [9]

Make “nominally feasible” and “robustly feasible” separate outputs.

Level 3 — B: How should reboost offset be reported?

Why this matters: a single percentage can conceal remaining spin, altitude, or scheduling deficits.

Answer: report transport-energy offset and angular-momentum recovery separately, then simulate electrodynamic control over the traffic schedule. An energy-only screening quantity can be

$$f_E = \frac{\sum E_{\text{recovered,in}}}{\sum E_{\text{expended,out}}},$$

but this is an analyst-defined metric—not a published asteroid-return reboost fraction. Use the simulated residual control requirement to report actual reboost reduction. Published HEFT modeling explicitly uses current control to boost the orbit or spin up the tether, demonstrating that those are distinct control objectives.^[1]

Level 3 — C: What if the catch fails?

Why this matters: an economical nominal trajectory can be unacceptable if a missed catch creates an uncontrolled Earth impact.

Answer: include a safe bypass or controlled-entry contingency, its targeting uncertainty, and the propellant needed to execute it. The Keck study makes safe destination and disposal part of mission design. OSIRIS-REx maintained a nonimpacting approach until final entry targeting

and used a spacecraft divert after capsule release. These are relevant operational precedents, not ready-made rotovator abort trajectories.^{[3] [9]}

A catch recommendation should therefore require a verified contingency as well as nominal rendezvous feasibility.

LOAD-BEARING NUMBERS

“Estimated” entries below are calculations from your design inputs or explicitly identified assumptions; their cited sources support the underlying published model or comparison datum, not the calculated result itself.

- 610 km — User-specified COM altitude; not a published MXER specification.
- 320 km — User-specified tether half-length; not a published MXER specification.
- 2.5 km/s — User-specified rotational tip speed.
- 7,301 km — User-specified upper-tip radius.
- $1.8 \text{ km}^2/\text{s}^2$ — Published 2008 HU4 arrival C_3 , 250 t variant.^[2]
- 250 t — Published returned mass for that 2008 HU4 variant.^[2]
- 1.342 km/s — Estimated excess speed from the published $1.8 \text{ km}^2/\text{s}^2$ arrival C_3 .^[2]
- $1.7 \text{ km}^2/\text{s}^2$ — Published 2008 HU4 arrival C_3 , 400 t variant.^[2]
- 400 t — Published returned mass for that 2008 HU4 variant.^[2]
- 1.304 km/s — Estimated excess speed from the published $1.7 \text{ km}^2/\text{s}^2$ arrival C_3 .^[2]
- $1.6 \text{ km}^2/\text{s}^2$ — Published arrival C_3 for the listed higher-mass 2008 HU4 variants.^[2]
- 650 t — Published 2008 HU4 returned-mass variant.^[2]
- 950 t — Published 2008 HU4 returned-mass variant.^[2]
- 1,300 t — Published 2008 HU4 returned-mass variant.^[2]
- 1.265 km/s — Estimated excess speed from the published $1.6 \text{ km}^2/\text{s}^2$ arrival C_3 .^[2]
- $2.0 \text{ km}^2/\text{s}^2$ — Published arrival C_3 for the listed KY26 and SG344 variants.^[2]
- 30 t — Published 1998 KY26 returned-mass variant.^[2]
- 60 t — Published 1998 KY26 returned-mass variant.^[2]
- 1,800 t — Published 2000 SG344 returned-mass variant.^[2]
- 1.414 km/s — Estimated excess speed from the published $2.0 \text{ km}^2/\text{s}^2$ arrival C_3 .^[2]
- $2.1 \text{ km}^2/\text{s}^2$ — Published 2000 SG344 arrival C_3 , 3,600 t variant.^[2]
- 3,600 t — Published 2000 SG344 returned mass for that variant.^[2]
- 1.449 km/s — Estimated excess speed from the published $2.1 \text{ km}^2/\text{s}^2$ arrival C_3 .^[2]
- $0.0 \text{ km}^2/\text{s}^2$ — Published manifold-capture variants; not an ordinary LEO parabolic-arrival specification.^[2]
- 12 km/s — Published NHATS maximum atmospheric-entry speed.^[7]
- 125 km — Published NHATS entry-speed evaluation altitude.^[7]

- 4.63 km/s — Estimated unbraked two-body excess-speed ceiling implied by the NHATS entry constraint. ^[7]
- 12.2 km/s — Published early OSIRIS-REx nominal entry velocity. ^[8]
- -8.2° — Published OSIRIS-REx entry flight-path angle; not asymptote declination. ^[8] ^[9]
- 12.38 km/s — Published operational OSIRIS-REx atmosphere-relative entry velocity. ^[9]
- 6,503.142 km — Published OSIRIS-REx entry-interface radius; used for the estimated comparison table. ^[9]
- 11.6 km/s — Published approximate Hayabusa2 geodetic/ground-relative entry velocity. ^[10]
- 5.12 km/s — Illustrative estimated excess speed assuming the early OSIRIS-REx entry velocity is inertial; not a reconstructed mission result. ^[8] ^[9]
- 10:1 — Published approximate MXER position-error-ellipse aspect ratio. ^[11]
- 5 s — Published upper bound on proposed Quadtrap closure duration. ^[11]
- 3.5 s — Published representative Quadtrap closure-design example. ^[11]
- 100 m — Published approximate maximum support-cable length in that closure example; not catch-position tolerance. ^[11]
- 2 g — Published representative MXER capture-mechanism acceleration used in that example. ^[11]
- 1:10 — Published approximate prototype dimensional scale. ^[11]
- 1 m — Published square prototype aperture side length. ^[11]
- 0.9 s — Published prototype passage-window duration. ^[11]
- 10 s — Published HASTOL capture-window upper bound, strictly shorter than this value. ^[12]
- 10,967 kg — Published control-station current-best-estimate mass without margin. ^[13]
- 8,274 kg — Published tether mass. ^[13]
- 650 kg — Published grapple mass. ^[13]
- 3,467 kg — Published retained-upper-stage mass. ^[13]
- 23,358 kg — Published initial facility on-orbit total. ^[13]
- 2,500 kg — Published initial payload mass in the cited facility designs. ^[1] ^[13]
- 0.107 — Estimated payload-to-facility ratio for the 23,358 kg configuration. ^[13]
- 9.34 — Estimated facility-to-payload ratio for that configuration. ^[13]
- 63 t — Published larger facility launch mass; not necessarily all-in on-orbit mass. ^[13]
- 5 t — Published larger point-design lunar/high-energy payload capability. ^[13]
- 10 t — Published larger point-design GTO payload capability. ^[13]
- 200 kW — Published larger point-design end-of-life reboost power. ^[13]
- One month — Published approximate reboost duration for that point design. ^[13]
- 26,250 kg — Published cislunar Earth-facility mass. ^[1]
- 10.5 — Published cislunar Earth facility-to-payload mass ratio. ^[1]

- 85 days — Published simulated cislunar facility orbit-restoration duration.^[1]
- 11 kW — Published power used for that reboost example.^[1]
- 95 days — Published architecture-level interval between lunar boosts.^[1]
- 15 days — Published LEO-GTO operational-orbit lifetime requirement.^[13]
- 30 days — Published LEO-GTO turnaround requirement.^[13]
- 1-5 km/s — User-requested arrival excess-speed comparison interval.
- 0.441 km/s — Estimated favorable-direction matching burn at arrival excess speed 1 km/s; user geometry and published energy-exchange model.^[1]
- 0.583 km/s — Estimated favorable-direction matching burn at 2 km/s.^[1]
- 0.815 km/s — Estimated favorable-direction matching burn at 3 km/s.^[1]
- 1.133 km/s — Estimated favorable-direction matching burn at 4 km/s.^[1]
- 1.528 km/s — Estimated favorable-direction matching burn at 5 km/s.^[1]
- 3.108 km/s — Estimated circular-capture burn at arrival excess speed 1 km/s.
- 3.250 km/s — Estimated circular-capture burn at 2 km/s.
- 3.483 km/s — Estimated circular-capture burn at 3 km/s.
- 3.800 km/s — Estimated circular-capture burn at 4 km/s.
- 4.195 km/s — Estimated circular-capture burn at 5 km/s.
- 11.117 km/s — Estimated direct-entry inertial speed at excess speed 1 km/s, using NASA's cited interface radius.^[9]
- 11.251 km/s — Estimated direct-entry inertial speed at excess speed 2 km/s.^[9]
- 11.471 km/s — Estimated direct-entry inertial speed at excess speed 3 km/s.^[9]
- 11.772 km/s — Estimated direct-entry inertial speed at excess speed 4 km/s.^[9]
- 12.149 km/s — Estimated direct-entry inertial speed at excess speed 5 km/s.^[9]
- 450 s — Analyst-selected propulsion specific-impulse assumption; not a cited mission specification.
- 50.6% — Estimated circular-capture propellant fraction at excess speed 1 km/s under that assumption.
- 52.1% — Estimated circular-capture propellant fraction at excess speed 2 km/s.
- 54.6% — Estimated circular-capture propellant fraction at excess speed 3 km/s.
- 57.7% — Estimated circular-capture propellant fraction at excess speed 4 km/s.
- 61.4% — Estimated circular-capture propellant fraction at excess speed 5 km/s.
- 10.449 km/s — Estimated escape speed at your upper-tip radius.
- 10.056 km/s — Estimated upper-tip inertial speed from your COM orbit and rotational speed.
- 0.393 km/s — Estimated minimum parabolic-arrival matching burn.
- 5.056 km/s — Estimated lower-tip inertial release speed.

- 43.03 MJ/kg — Estimated upper-to-lower payload energy transfer in the negligible-recoil approximation; published model basis. ^[1]
- 5.336 km/s — Estimated lower-tip-release entry speed at the cited interface radius. ^[9]
- 22% — NASA-published Stardust heatshield TPS fraction of entry mass. ^[15]
- 18% — NASA-published Genesis heatshield TPS fraction of entry mass. ^[15]
- 14% — NASA-published Apollo heatshield TPS fraction of entry mass. ^[15]
- 12.5 km/s — NASA-published approximate Stardust entry speed. ^[15]
- 1,200 W/cm² — NASA-published approximate Stardust peak stagnation heating. ^[15]
- 12 MW/m² — Mission-author-published approximate Hayabusa2 maximum heat flux. ^[10]
- 1,051 W/cm² — Published OSIRIS-REx peak-heat-rate survivability limit, not nominal measured peak. ^[9]
- 30.3 kJ/cm² — Published OSIRIS-REx integrated-heat-load survivability limit. ^[9]
- 40 g — Published OSIRIS-REx peak-deceleration survivability limit. ^[9]
- 99% — Published required Monte Carlo compliance fraction for those OSIRIS-REx survivability limits. ^[9]
- 10–16 km/s — Published Earth-entry TPS estimating-study velocity range. ^[14]
- 5–25° — Published absolute entry-flight-path-angle range in that study. ^[14]
- 41.95–128.74 kg/m² — Published ballistic-coefficient range in that study. ^[14]
- 1,200 W/cm² — Published cold-wall heating applicability limit for that study's PICA correlation. ^[14]
- 6.3% — Published PICA-correlation standard-deviation agreement with FIAT within the evaluated dataset. ^[14]
- 11.7% — Published maximum PICA-correlation underprediction within the evaluated dataset. ^[14]

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